Conversion of direct form-I structure to direct form-II structure

The direct form-I structure can be converted to direct form-II structure by considering the direct form-I structure as cascade of two systems $H_1(z)$ and $H_2(z)$ as shown in Figure 4.8(a). By linearity property, the order of cascading can be interchanged as shown in Figure 4.8(b).

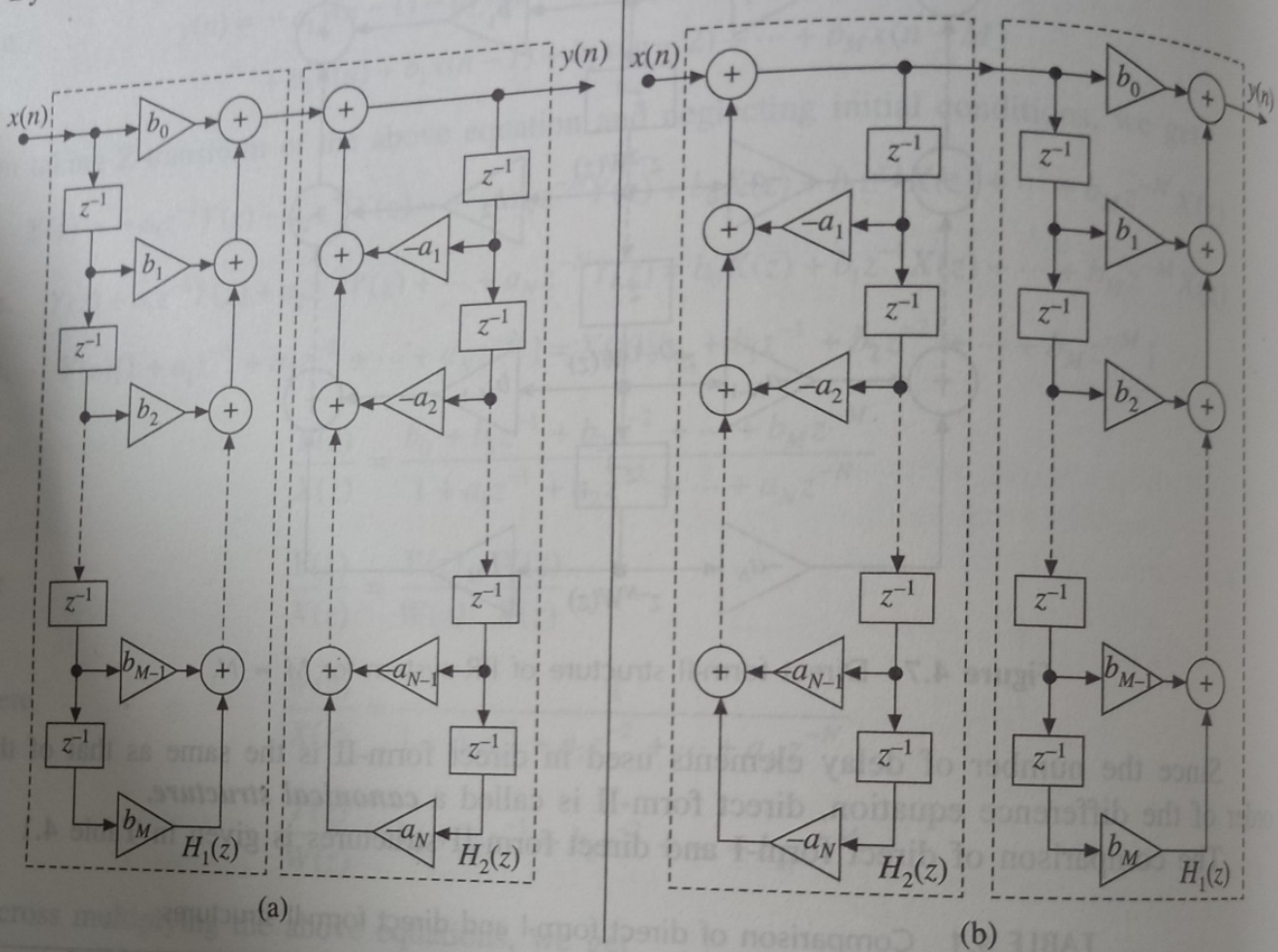

Figure 4.8 (a) Direct form-I structure as cascade of two systems (b) Direct form-I structure after interchanging the order of cascading.

In Figure 4.8(b), we can observe that the inputs to the delay elements in $H_1(z)$ and $H_2(z)$ are the same and so the outputs of the delay elements in $H_1(z)$ and $H_2(z)$ are same. Therefore, instead of having separate delays for $H_1(z)$ and $H_2(z)$, a single set of delays can be used. Hence, the delays can be merged to combine the cascaded systems to a single system. The resultant structure will be direct form-II structure as that of Figure 4.7. The process of converting direct form-I structure to direct form-II structure is shown in Figure 4.9.

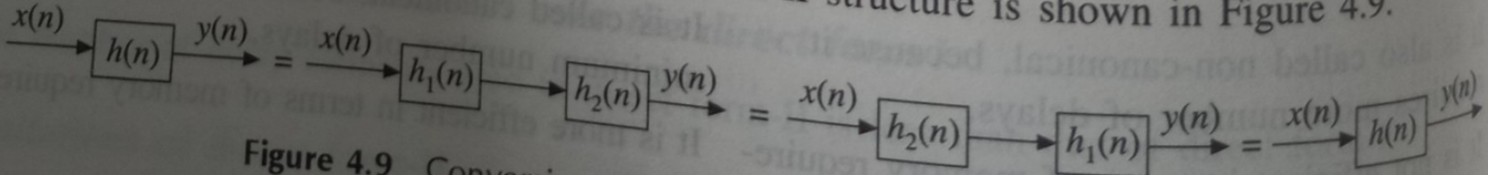

Figure 4.9 Conversion of direct form-I structure to direct form-II.

EXAMPLE 4.2 Realize an IIR system

$$y(n) + 2y(n-1) + 3y(n-2) = 4x(n) + 5x(n-1) + 6x(n-2)$$

using the transposed form structure.

Solution: Taking Z-transform on both sides of the given difference equation and neglecting initial conditions, we get

$$Y(z) + 2z^{-1}Y(z) + 3z^{-2}Y(z) = 4X(z) + 5z^{-1}X(z) + 6z^{-2}X(z)$$

Therefore, the transfer function of the given IIR system is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{4 + 5z^{-1} + 6z^{-2}}{1 + 2z^{-1} + 3z^{-2}}$$

Figure 4.10 (a) General transposed structure realization of IIR system through direct form-I, (b) General transposed structure realization of IIR system through direct form-II.

The direct form-II realization structure, the recovered realization structure and the transposed form realization structure of this system are shown in Figure 4.11[(a), (b) and (c) respectively].

$$a_1 = 2, a_2 = 3, b_0 = 4, b_1 = 5, b_2 = 6$$

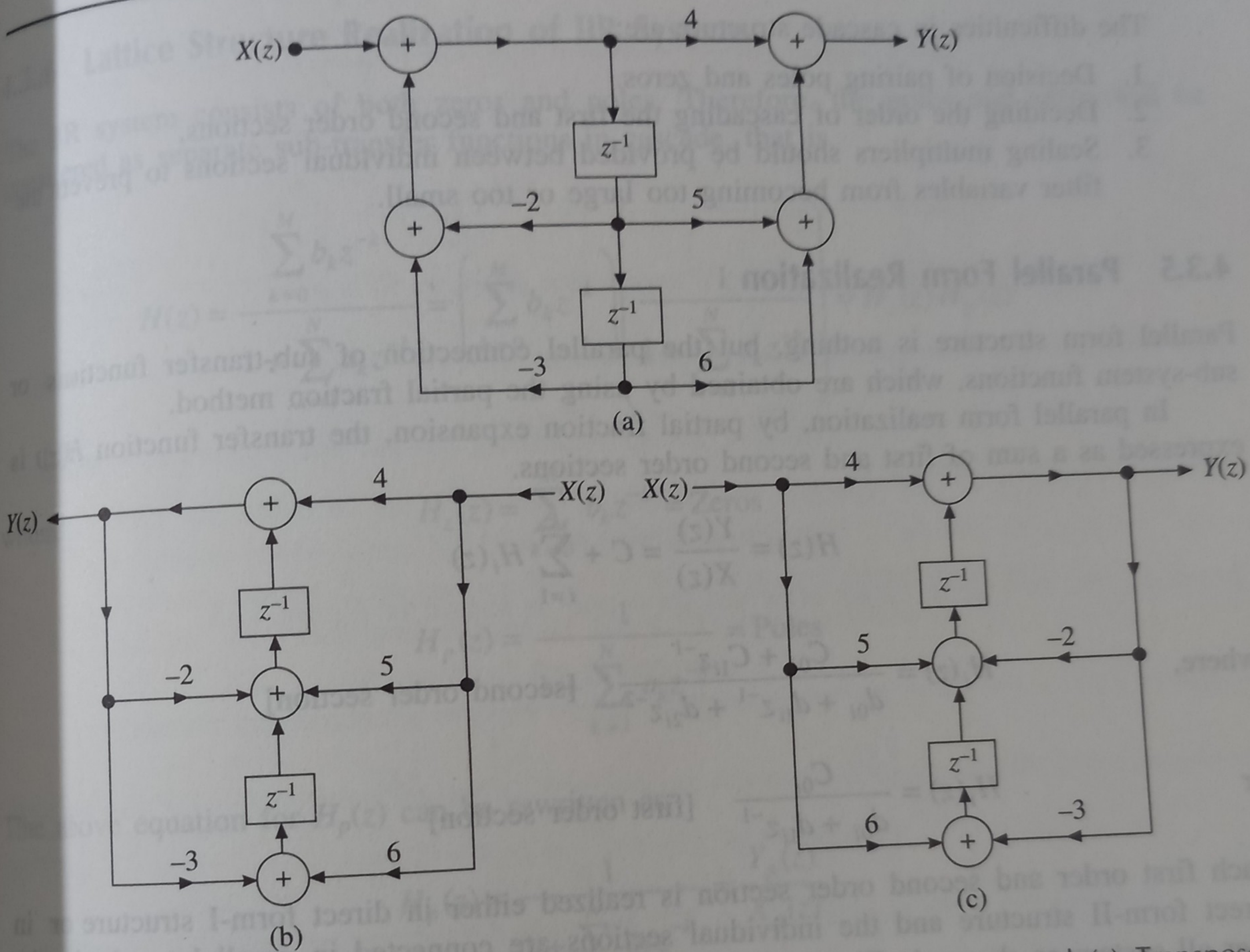

Figure 4.11 (a) Direct form-II realization (b) Recovered realization structure and (c) Transposed realization structure (Example 4.2).